

Formula sheet for Quiz 3

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more formulas than any one question needs, so deciding which applies is still your job.

A layer, and the forward pass

$$\begin{array}{lll} \text{one neuron} & z = \mathbf{w}^\top \mathbf{x} + b, & a = \sigma(z) \\ \text{a layer} & \mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}, & \mathbf{a} = \sigma(\mathbf{z}) \\ \text{the whole network} & \mathbf{a}^{(0)} = \mathbf{x}, \quad \mathbf{a}^{(\ell)} = \sigma^{(\ell)}(\mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}), & \hat{\mathbf{y}} = \mathbf{a}^{(L)} \end{array}$$

$\mathbf{W}$ is $(n_{\text{out}} \times n_{\text{in}})$, one row per unit, and $\mathbf{b}$ is (n_{out}) , so one layer holds $n_{\text{out}} \times n_{\text{in}} + n_{\text{out}}$ parameters. The superscript is a **layer index**, not a power, so a network is a composition: $\hat{y} = f_3(f_2(f_1(\mathbf{x})))$.

Activations

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad \text{ReLU}(z) = \max(0, z)$$

Losses

$$\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 \quad \text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \quad \text{BCE} = -\frac{1}{n} \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

The chain rule

$$\frac{d}{dx} [f(g(h(x)))] = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Multivariable version. If u reaches the output along several intermediate quantities $v_1, \dots, v_k$:

$$\frac{\partial \mathcal{L}}{\partial u} = \sum_k \frac{\partial \mathcal{L}}{\partial v_k} \frac{\partial v_k}{\partial u}$$

Gradient descent, and training vocabulary

$$\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} - \lambda \nabla \mathcal{L}(\mathbf{w}_{\text{old}}) \quad \text{iterations per epoch} = \left\lceil \frac{N}{B} \right\rceil$$

Batch size B : examples per update. **Iteration**: one parameter update. **Epoch**: one full pass through the training set. **Learning rate** λ : scales every parameter's step.

Finite differences

$$\frac{\partial \mathcal{L}}{\partial w_j} \approx \frac{\mathcal{L}(w_j + \varepsilon) - \mathcal{L}(w_j)}{\varepsilon}$$

Nudge one parameter by ε , re-run the forward pass, see how much the loss moved, put it back.

Notation

$\nabla_{\mathbf{x}} F$ is the gradient of F with respect to $\mathbf{x}$; θ is the full set of network parameters; ℓ is a single example's loss and $\mathcal{L}$ is the cost; J appears in place of $\mathcal{L}$ in some sources.